Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Differentiable and Robust Contact-Aware Planning Technology for Scalable Robotic Manipulation**				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Mechanical · Materials			Mechanical · Materials	
	Medium Category		Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery		Robot Control and Intelligence Technologies	*Robotics/ Automation Machinery*	Mechanical Automation Technologies
Concept of Proposed Technology	The proposed technology develops a **differentiable, contact-aware planning and control framework** that enables robots to efficiently generate robust manipulation policies in contact-rich environments. Robot learning today relies heavily on teleoperated demonstrations and sampling-based optimization (e.g., RL, MPPI, CEM). These approaches are costly, sample-inefficient, and difficult to scale—especially for manipulation tasks involving contact, friction, and deformable objects. Our core innovation is a **differentiable contact simulation and robust trajectory optimization framework** that:				

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	• Smooths hybrid contact dynamics to provide informative gradients • Quantifies the discrepancy between smoothed and true dynamics • Synthesizes robust state-feedback and output-feedback controllers • Enables scalable simulation-based data generation • Supports multimodal sensing integration (vision, force, tactile) The proposed technology develops a **differentiable contact simulation framework** that enables gradient-based planning for contact-rich robotic tasks. By smoothing complementarity-based contact models and differentiating the resulting KKT conditions, the simulator provides continuous and informative gradients across contact modes, allowing efficient first-order optimization methods to be applied to manipulation problems. Building on this, we formulate a **robust contact-aware trajectory optimization framework** that accounts for the bounded modeling error introduced by smoothing. By analytically characterizing this discrepancy and embedding it into a robust optimization problem, we synthesize feedback controllers that guarantee constraint satisfaction and reliable performance under true contact dynamics. Finally, we develop **multimodal output-feedback policy synthesis** that leverages tactile, force, and vision signals for robust control. By explicitly handling sensing noise and deploying policies across diverse simulated environments, the framework enables scalable multimodal data generation and reduces reliance on costly real-world demonstrations. This framework reduces dependence on costly real-world teleoperation and enables industrial partners to generate high-quality manipulation policies efficiently.
Rationale for Joint R&D Project	Successful commercialization requires tight integration between: • Advanced control and differentiable simulation algorithms (research institute) • Industrial robotic hardware platforms, sensing systems, and embedded software (Korean SME) Collaboration is essential because: • Contact-rich policy synthesis must be validated on real robotic hardware • Sensor noise, actuator limits, and embedded constraints must be incorporated into the optimization loop • Robustness guarantees must be experimentally verified under industrial conditions Expected benefits of joint R&D: • Reduced data collection cost for manipulation policy training

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	• Faster deployment of contact-robust robotic systems • Transferable software stack applicable to manufacturing, logistics, and service robotics • Strengthened global competitiveness of Korean robotics SMEs
Key Words	1. Differentiable Contact Simulation 2. Robust Trajectory Optimization 3. Contact-Rich Manipulation 4. Multimodal Sensor Feedback Control 5. Scalable Robot Data Generation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop and validate a **robust, differentiable contact-aware planning framework** integrated into an industrial robotic manipulation platform. The specific R&D objectives are to: • Develop differentiable contact solvers with informative gradients • Derive analytical robustness bounds for smoothed dynamics • Implement robust trajectory optimization and feedback synthesis • Integrate multimodal sensing (force/tactile/vision) • Demonstrate scalable simulation-to-real transfer on contact-rich tasks			
R&D Target Specifications	**The target performance specifications for this R&D project are:** • **Contact Planning Success Rate (≥ 85%).** We aim to achieve ≥ 85% success rate on representative contact-rich manipulation tasks (pushing, insertion, deformable interaction) under real hardware deployment. • **Constraint Satisfaction Reliability (≥ 99%).** We aim to maintain ≥ 99% safety constraint satisfaction (joint limits, collision avoidance, force limits) under bounded modeling error. • **Simulation-to-Real Transfer Gap (< 10%).** We aim to limit performance degradation from simulation to hardware deployment to less than 10%. • **Data Generation Efficiency (≥ 3× improvement).** We aim to reduce required real-world data collection by at least 3× compared to sampling-based baselines.			
R&D Plan	This R&D project will be conducted over three years, with a detailed plan as follows: **Year 1: Differentiable Contact Simulation & Robust Modeling** • Develop smoothed complementarity-based contact solver • Implement implicit differentiation through KKT systems • Derive analytical error bounds between true and smoothed dynamics • Validate gradient quality and stability in simulation • Benchmark against sampling-based baselines **Deliverable**: Differentiable contact simulation engine (lab validated) **Year 2: Robust Policy Synthesis & Hardware Integration** • Formulate and solve robust trajectory optimization problems • Develop state-feedback and output-feedback controller synthesis • Integrate algorithms with SME's robotic manipulation platform • Incorporate multimodal sensing (force/tactile) • Conduct controlled laboratory experiments **Deliverable**: Integrated manipulation prototype (laboratory validation)			

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	Year 3: Pilot-Scale Validation & Industrial Demonstration • Deploy system in industrially relevant scenarios • Perform large-scale robustness and safety validation • Optimize computational performance for real-time execution • Refine commercialization roadmap and IP strategy **Deliverable**: Pilot-scale validated prototype (TRL 6)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the core AI, control, and verification algorithms and intelligence for the proposed manipulation technology. Specifically, the research institute will: • Develop differentiable contact models and smoothing strategies • Derive robustness guarantees and analytical bounds • Design and implement robust trajectory optimization algorithms • Develop multimodal output-feedback policy synthesis methods • Provide theoretical analysis and performance benchmarking • Support system integration and technical validation
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the robotic hardware and system-level implementation required for the proposed home mobile manipulation technology. The SME will play a key role in translating AI algorithms into a practical and deployable robotic system. Specifically, the partner will: • Provide industrial robotic manipulation platform • Integrate sensing hardware (force, tactile, vision) • Implement embedded software and real-time interfaces • Conduct laboratory and pilot-scale experiments • Provide industrial test scenarios and validation metrics • Lead commercialization planning and product integration

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**The anticipated technical impact is the following:** • First industrial-grade differentiable contact planning framework with robustness guarantees • Efficient generation of contact-rich manipulation policies • Reduced reliance on teleoperation and sampling-based training • Validated sim-to-real pipeline for multimodal robotic control **The anticipated industrial impact is the following:** • Enables Korean SMEs to reduce data acquisition cost • Accelerates deployment of manipulation robots in manufacturing and logistics • Improves robustness and safety of industrial robotic systems • Establishes a competitive software-control IP portfolio **The anticipated broader impact is the following:** • Advances Korea's competitiveness in AI-driven robotics • Creates foundation technology for next-generation intelligent manufacturing • Enables scalable training of contact-rich robotic systems

6. References

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(3) Glen Chou, Russ Tedrake. "Synthesizing Stable Reduced-Order Visuomotor Policies for Nonlinear Systems via Sums-of-Squares Optimization," in *IEEE Conference on Decision and Control (CDC), Singapore, 2023.*

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(5) Hyung Ju Suh, Glen Chou, Hongkai Dai, Lujie Yang, Abhishek Gupta, Russ Tedrake. "Fighting Uncertainty with Gradients: Offline Reinforcement Learning via Diffusion Score Matching," in *Conference on Robot Learning*, Atlanta, GA, 2023.,

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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